

Week 6 · Student Notes

AI531C Digital Image Analysis · Tue 8 – Wed 9 Sep 2026 · Unit II · Lectures 14, 15, 16

Everything taught in this week, in one document. The per-lecture handouts are identical in content — use whichever you prefer.

What is in this bundle

- Lecture 14 — Image Registration, Alignment and Homography
- Lecture 15 — Edge Detection and Contour Extraction
- Lecture 16 — Thresholding, Region Methods and Watershed Segmentation

This week at a glance

Sessions — Tue 8 – Wed 9 Sep 2026.

Tutorial — T6 (Tue 8 Sep).

Pre-class preparation — PCM-6.

Released after the last class of the week: Wed 9 Sep 2026, 18:00.

How to use this bundle — read the lecture section on the evening of the class while the argument is fresh, attempt the check-your-understanding questions before the tutorial, and leave the practice questions until you can do them unaided. No answer keys are printed: the reasoning is worked through in the tutorial.

Lecture 14 · Image Registration, Alignment and Homography

AI531C Digital Image Analysis · Unit II · CO2 · M.Tech. Artificial Intelligence, Department of CSE, DTU

Finding the transform nobody gave you — features, matches, liars, and the algorithm that outvotes them

What you should already know

This lecture continues from Lecture 13, “Geometric Transformations and Interpolation”. Revise that material first — today's arguments assume it.

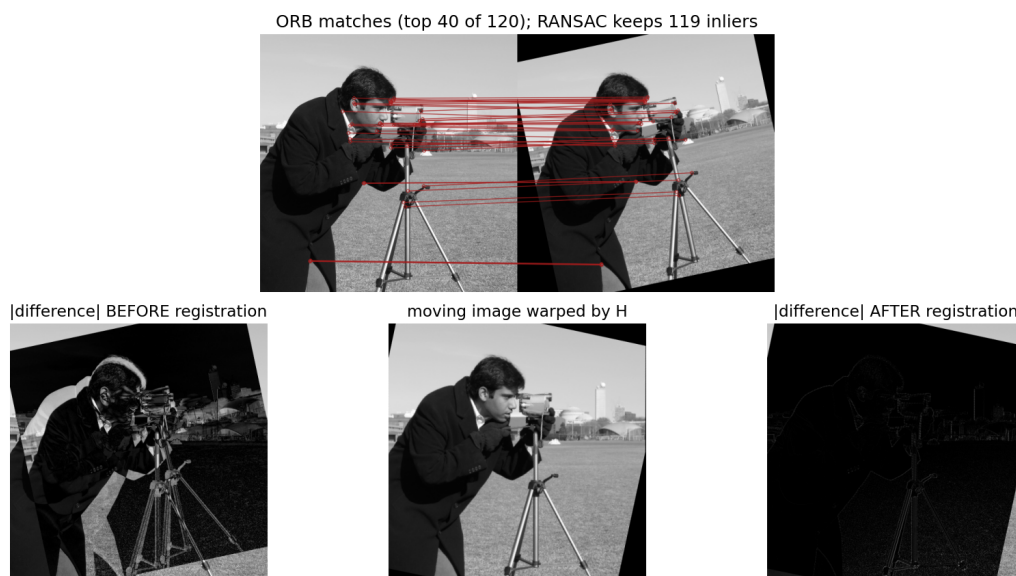
Reading in the course notes: Comprehensive Notes L14; Vol. I ch. 14.

Learning outcomes

After studying this lecture you should be able to:

- state the registration pipeline: detect $\rightarrow$ describe $\rightarrow$ match $\rightarrow$ estimate $\rightarrow$ warp.
- explain feature-based vs intensity-based alignment and when each wins.
- estimate a homography from matches and reject outliers with RANSAC.
- compute the RANSAC iteration count.
- evaluate registration with reprojection error and difference images.

Figure 1: The whole pipeline on one pair



ORB matches, RANSAC, warp, and difference images before/after. Generated by L14_registration_homography.py.

How to read this figure

What to notice — 120 candidate matches; RANSAC keeps 119. Mean |difference| falls 43.6 $\rightarrow$ 13.1 after warping.

Why it happens — Distinctive corners match reliably here; the single outlier would still have poisoned a least-squares fit.

What it means — The difference image is registration's report card — structure remaining = misalignment (or actual change!).

Why it matters — Never difference unregistered images; never trust matches without a robust fit.

Core concepts

The five-stage pipeline

- 1 · Detect — Find repeatable interest points (corners, blobs) in both images — locations that survive viewpoint change.
- 2 · Describe — Summarise each point's neighbourhood as a vector (ORB: binary tests; robust to rotation/illumination to a degree).
- 3 · Match — Nearest-neighbour in descriptor space + cross-check; some matches WILL be wrong — plan for liars.
- 4 · Estimate robustly — RANSAC: fit homographies to random minimal sets (4 matches), keep the one most matches agree with.
- 5 · Warp and verify — Apply H (L13 machinery); judge by reprojection error and the difference image.

Homography estimation and its error

Problem: from N matched pairs ($x_i \leftrightarrow x'_i$), find the 3×3 H mapping plane-to-plane. Scene planar or camera rotating (homography validity conditions — state them!).

$$x'_i \approx \pi(H x_i) \quad e_i = \|x'_i - \pi(H x_i)\|$$

What each symbol means

- x, x' — homogeneous points in image 1, 2 π — perspective division by w (L13) e_i — reprojection error of match i (pixels) Inlier: $e_i < \tau$ (threshold, e.g. 3 px)

How to read this

What it means — With $N \geq 4$ the system is overdetermined — good, because redundancy exposes liars. Median/inlier-mean reprojection error is THE registration quality number to report.

Worked example

Given — Half the candidate matches are wrong (inlier ratio $w = 0.5$). A homography needs a minimal sample of $s = 4$ matches. We want 99% confidence ($p = 0.99$) of drawing at least one all-inlier sample.

Find — The required number of RANSAC iterations N .

Formula — $N = \log(1 - p) / \log(1 - w^s)$

Solution

Substitution — $w^4 = 0.0625$; $N = \log(0.01) / \log(0.9375)$

Calculation — $N = (-4.605) / (-0.0645) = 71.4$ ANSWER: $N = 72$ iterations

What it means — 72 random tries beat 50% liars with 99% confidence — cheap. But N explodes as w falls: at $w = 0.2$, $N \approx 2\,876$. Cleaning matches BEFORE RANSAC is never wasted work.

The code behind the results

```
orb = cv2.ORB_create(1000)
k1, d1 = orb.detectAndCompute(im1, None)
k2, d2 = orb.detectAndCompute(im2, None)
m = cv2.BFMatcher(cv2.NORM_HAMMING, crossCheck=True).match(d1, d2)
H, mask = cv2.findHomography(p2, p1, cv2.RANSAC, 3.0)
aligned = cv2.warpPerspective(im2, H, wh)
```

Parameters used — 1000 features; Hamming + crossCheck; $\tau = 3$ px. Full program: L14_registration_homography.py

What it means — mask marks the inliers — count them; report the ratio.

Something to think about — $\tau = 0.5$ px? Inliers starve, H unstable. $\tau = 20$ px? Liars sneak in. The threshold is a modelling decision.

Feature-based vs intensity-based

Aspect	Feature-based (today)	Intensity-based (direct)
Works from	Sparse distinctive points	All pixels via a similarity metric
Large displacement	Handles it (global matching)	Fails without a good initial guess
Low-texture images	Starves (no corners)	Wins (medical, smooth scenes)
Robustness path	RANSAC over matches	Robust metrics + multi-scale (L9 pyramids!)
Cost profile	Cheap once features exist	Iterative optimisation per pair

Key takeaway: texture-rich + big motion → features; smooth + small motion → intensity. Hybrid (features initialise, intensity refines) is common practice.

Where this is used — your project

Change-detection lane (critical path) — registration error masquerades as change — report inlier count, inlier ratio and mean reprojection error BEFORE any change map; your L38 deep models inherit whatever misregistration survives;

Fusion lane — multi-sensor pairs must be registered before any fusion rule (L40); cross-modality often defeats ORB — plan mutual-information-based alignment or manual control points, and SAY which;

All lanes — the difference image after alignment is a free diagnostic — archive it for every registered pair.

Try this yourself — Register your PCM-6 pair tonight with the lecture code; save the before/after difference images for T6.

Summary

- Registration = detect, describe, match, estimate robustly, warp, VERIFY. Redundant correspondences + RANSAC turn lying matches into a minority vote. The aligned difference image is the report card — and the doorway to change detection.

Key formulae

$$e_i = \|x_i' - \pi(H x_i)\|; \text{ inlier: } e_i < \tau N = \log(1-p)/\log(1-w^s) \text{ Homography valid: planar scene or pure rotation}$$

What comes next

Tutorial T6 — today 2:00–3:00 PM (studio) — Geometry & registration studio: run the full pipeline on YOUR PCM-6 pair; artefact = registered pair + inlier stats + before/after difference images + one mismatch explained.

Before tomorrow (L15–L16 double) — Nothing due. PCM-7 (segmentation design memo) releases Friday — it needs L15/L16 content, so attend first.

Logistics — L15 (Wed 10 AM): edge detection. L16 (Wed 11 AM): thresholding & watershed. T7 (Tue 15 Sep) doubles as the project-proposal clinic.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Find the liar

Your PCM-6 pair produced 6 hand-marked correspondences. Five agree on 'shift right ~40 px, slight rotation'. One maps a window corner to a similar-looking corner TWO windows over. 1. What happens to a least-squares homography fit that includes all 6? 2. What does RANSAC do with match #6, mechanically? 3. Why did the mismatch happen at a WINDOW, of all places?

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, understand): List the five stages of feature-based registration and state where outliers enter and where they are removed.

Q2 (6 marks, apply): With inlier ratio 0.6 and confidence 0.99, compute RANSAC iterations for homography estimation ($s = 4$). Repeat for affine ($s = 3$) and explain the difference.

Q3 (10 marks, analyse): A team registers drone images of a hilly (non-planar) field with a single homography; the centre aligns well but edges show systematic ghosting. Diagnose via the homography's validity conditions and propose two remedies.

Further reading

Szeliski Ch.6 & Ch.9; Forsyth-Ponce robust alignment

Course notes: Comprehensive Notes L14; Vol. I ch. 14.

Runnable code for every figure and number in this lecture: 06_CODE_AND_DEMOS (see the file names quoted beside each figure).

AI531C Digital Image Analysis · student handout · every figure and numerical result in this document is the output of course code you can run yourself.

Lecture 15 · Edge Detection and Contour Extraction

AI531C Digital Image Analysis · Unit II · CO2 · M.Tech. Artificial Intelligence, Department of CSE, DTU

Where images change fastest — gradients, Laplacians, and why Canny's three-stage discipline still sets the standard

What you should already know

This lecture continues from Lecture 14, “Image Registration, Alignment and Homography”. Revise that material first — today's arguments assume it.

Reading in the course notes: Comprehensive Notes L15; Vol. I ch. 15.

Learning outcomes

After studying this lecture you should be able to:

- compute image gradients with Sobel/Prewitt and read magnitude + direction.
- contrast first-derivative and Laplacian edge evidence.
- explain Canny's three stages and the role of each.
- predict the effect of σ and the hysteresis thresholds.
- diagnose noisy, broken and doubled edge maps.

The gradient: direction and strength of change

Problem: quantify local change at every pixel. Approximate partial derivatives with small kernels; assume mild pre-smoothing (noise fights derivatives — quantified today).

$$|\nabla f| = \sqrt{(f_x)^2 + (f_y)^2} \quad \theta = \text{atan2}(f_y, f_x) \quad \nabla^2 f = f_{xx} + f_{yy}$$

What each symbol means

- f_x, f_y — Sobel/Prewitt responses (horizontal, vertical) $|\nabla f|$ — edge strength; θ — edge NORMAL direction $\nabla^2 f$ — Laplacian: second derivative, zero-crossing at edges Sobel = smoothing $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \times$ derivative $\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$ — separable!

How to read this

What it means — First derivative: edges are MAXIMA of $|\nabla f|$. Second derivative: edges are ZERO-CROSSINGS of $\nabla^2 f$ — thinner localisation, far touchier about noise. Differentiation amplifies high frequencies (L9 view: derivative = ramp filter) — hence smoothing first, always.

Worked example

Given — 3×3 patch spanning a vertical step: every row is [10, 10, 80]. Sobel kernels: $G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$, $G_y = G_x^T$.

Find — f_x, f_y , gradient magnitude and direction at the centre.

Formula — $f_x = \sum G_x \circ \text{patch}$; $|\nabla f| = \sqrt{(f_x)^2 + (f_y)^2}$; $\theta = \text{atan2}(f_y, f_x)$

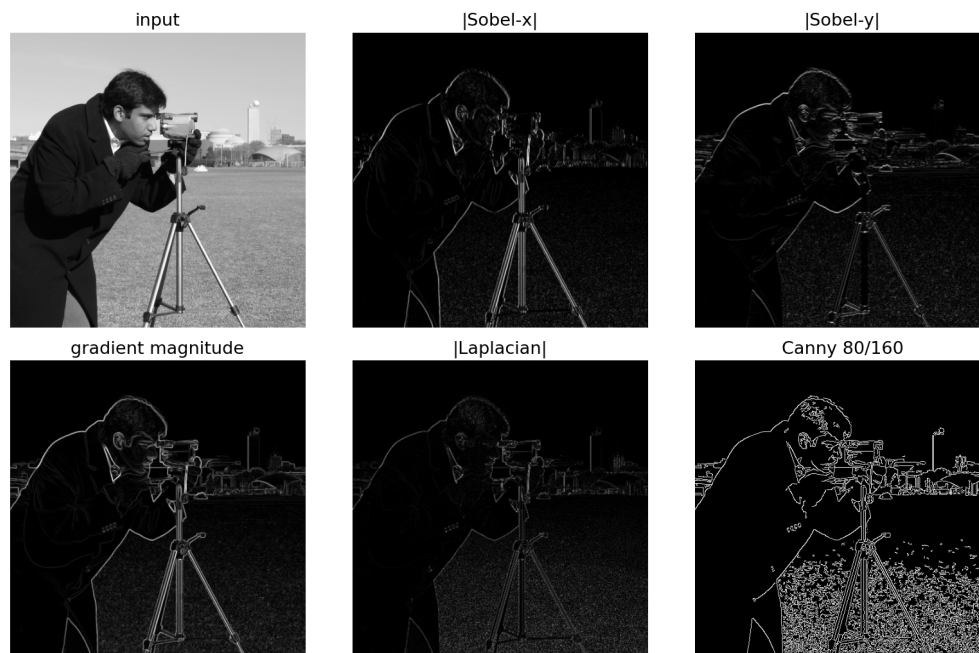
Solution

Substitution — $f_x = (-10 - 20 - 10) + (80 + 160 + 80) = -40 + 320$

Calculation — $f_x = 280$; $f_y = 0$ (rows identical) ANSWER: $|\nabla f| = 280$, $\theta = 0^\circ$ (gradient points along +x)

What it means — θ is the direction of steepest INCREASE — perpendicular to the edge line itself. The factor-of-4 weighting (2s in the middle row) is Sobel's built-in smoothing at work.

Figure 1: The operator gallery



Sobel components, magnitude, Laplacian, Canny on camera(). Generated by L15_edge_detection.py.

How to read this figure

What to notice — |Sobel-x| lights vertical structure; |Sobel-y| horizontal; magnitude merges both; Laplacian is thin but busy; Canny is thin AND clean.

Why it happens — Directional kernels see their own orientation; the Laplacian's second derivative doubles noise sensitivity; Canny post-processes the gradient.

What it means — Raw operators give EVIDENCE; Canny turns evidence into a decision.

Why it matters — Downstream tasks (L16 contours) want decisions — hence Canny's ubiquity.

The code behind the results

```
sx = cv2.Sobel(img, cv2.CV_32F, 1, 0)
sy = cv2.Sobel(img, cv2.CV_32F, 0, 1)
mag = np.sqrt(sx**2 + sy**2)
edges = cv2.Canny(blurred, 80, 160)
# CV_32F: keep negative gradients!
```

Parameters used — $T_{\text{low}}/T_{\text{high}} \approx 1:2$ to $1:3$; σ via prior GaussianBlur. Full program: L15_edge_detection.py

What it means — Canny wants the RAW image + your smoothing choice — don't double-smooth.

Something to think about — uint8 Sobel? Negative responses clip to 0 — half of every edge vanishes (L8's exam trap, live).

Choosing the edge tool

Tool	Evidence type	Strength	Weakness
Sobel/Prewitt	1st derivative, directional	Cheap; gives direction; robust-ish	Thick responses; threshold needed
Laplacian/LoG	2nd derivative, isotropic	Thin zero-crossing localisation	Noise-doubled; no direction
Canny	Gradient + NMS + hysteresis	Thin, linked, tunable — the default	Three parameters to justify
Gradient magnitude alone	Raw evidence	Feeds custom pipelines (watershed L16!)	Not a decision — needs a consumer

Key takeaway: pick by what consumes the output — decisions want Canny; algorithms (watershed, snakes) often want raw magnitude.

Where this is used — your project

Segmentation lane — edge maps are boundary evidence for your classical baseline — and edge-recall against ground truth is a defensible custom metric;

Deblurring / super-resolution lanes — edge sharpness is your qualitative exhibit A — show the same edge profile before/after as a 1-D plot (judges love it, rubric rewards it);

Change-detection lane — edge-difference maps resist illumination drift better than intensity differences — a free robustness upgrade for the L18 baseline.

Try this yourself — Canny threshold sweep on one project image; save the three maps — instant appendix figure.

Summary

- Gradients give strength AND direction; direction is normal to the edge. Derivatives amplify noise — smoothing first is part of the operator, not an option. Canny = evidence → thinning → connected decision; its parameters are claims to justify.

Key formulae

$|\nabla f| = \sqrt{(f_x)^2 + (f_y)^2}$; $\theta = \text{atan2}(f_y, f_x)$ Sobel = $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}^T \otimes \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$ Hysteresis: keep weak only if 8-connected to strong

What comes next

Immediately next (L16, same room) — Thresholding, region methods and watershed — today's gradient magnitude becomes the watershed's landscape.

PCM-7 (releases Fri 11 Sep, due Mon 14 Sep, ~35 min) — Segmentation design memo: propose thresholding + edge + morphology stages for a supplied uneven-illumination, touching-objects problem — one pipeline diagram.

Logistics — T7 (Tue 15 Sep): segmentation & morphology studio + project-proposal clinic. PCM-7 pipelines get defended in teams at T7.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Predict the sweep

Same noisy image, three Canny settings (see panel): A. No smoothing, thresholds 80/160. B. $\sigma = 1.4$ smoothing, 80/160. C. $\sigma = 1.4$, thresholds 20/40. Predict each output in one phrase BEFORE looking: which is speckled, which is clean, which is overgrown?

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, understand): Why does Canny apply non-maximum suppression along the gradient direction, and what artefact appears without it?

Q2 (6 marks, apply): For a 3×3 patch with columns [50, 50, 130] in every row, compute Sobel f_x , f_y , $|\nabla f|$ and θ , and state the Prewitt magnitude for comparison.

Q3 (10 marks, analyse): A PCB inspection line uses Canny 100/200 tuned in summer; in winter (drier air, different lighting) breaks appear in traces on the edge map. Diagnose two plausible mechanisms and design a robust re-parameterisation.

Further reading

GW4 Ch.10 Sec.10.2; Szeliski Ch.4; Forsyth-Ponce edges

Course notes: Comprehensive Notes L15; Vol. I ch. 15.

Runnable code for every figure and number in this lecture: 06_CODE_AND_DEMOS (see the file names quoted beside each figure).

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Lecture 16 · Thresholding, Region Methods and Watershed Segmentation

AI531C Digital Image Analysis · Unit II · CO2 · M.Tech. Artificial Intelligence, Department of CSE, DTU

From boundaries to regions: letting the histogram choose the threshold, and letting water separate what touches

What you should already know

This lecture continues from Lecture 15, “Edge Detection and Contour Extraction”. Revise that material first — today's arguments assume it.

Reading in the course notes: Comprehensive Notes L16; Vol. I ch. 16.

Learning outcomes

After studying this lecture you should be able to:

- apply global, Otsu and adaptive thresholding to matched conditions.
- derive Otsu's criterion and compute it on a small histogram.
- explain region growing and its seed/criterion decisions.
- run marker-based watershed and control over-segmentation.
- assemble a classical segmentation pipeline for a stated problem.

Worked example

Given — 4-level image, histogram counts $n = [4, 2, 1, 3]$ ($N = 10$ pixels), levels $k = 0 \dots 3$.

Find — $\sigma_B^2(t)$ for $t = 0, 1, 2$ and the Otsu threshold.

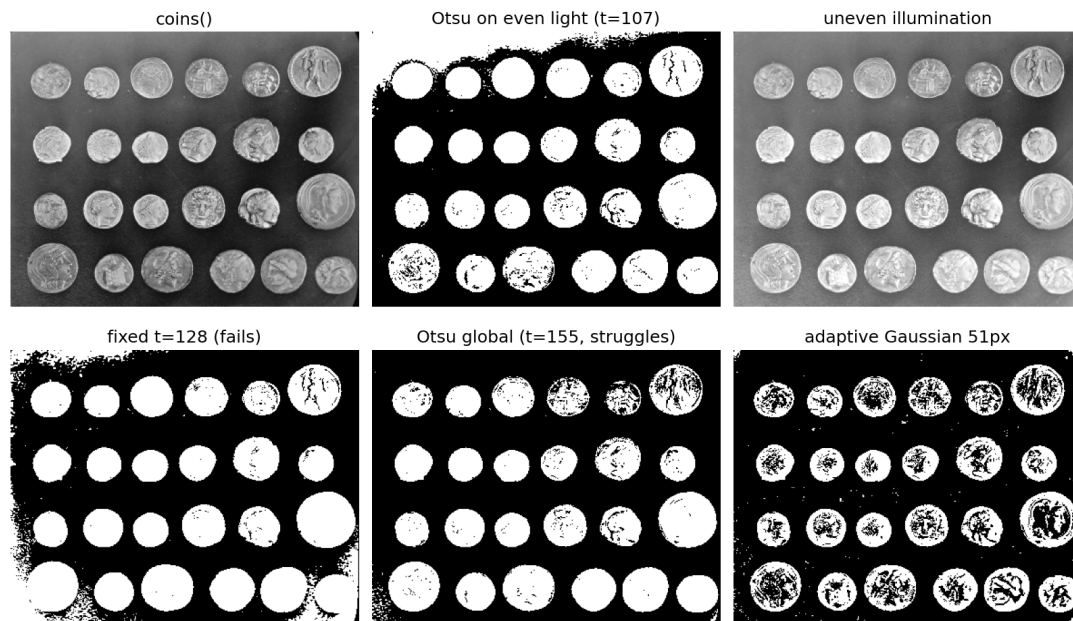
Formula — $\sigma_B^2(t) = \omega_0 \omega_1 (\mu_0 - \mu_1)^2$

Solution

Working — $t=0$: $\omega_0=0.4$, $\mu_0=0$, $\mu_1=2.167 \rightarrow \sigma_B^2 = 0.4 \cdot 0.6 \cdot 2.167^2 = 1.127$ $t=1$: $\omega_0=0.6$, $\mu_0=0.333$, $\mu_1=2.750 \rightarrow \sigma_B^2 = 0.6 \cdot 0.4 \cdot 2.417^2 = 1.402$ $t=2$: $\omega_0=0.7$, $\mu_0=0.571$, $\mu_1=3.0 \rightarrow \sigma_B^2 = 0.7 \cdot 0.3 \cdot 2.429^2 = 1.239$ ANSWER: $t^* = 1$ ($\sigma_B^2 = 1.402$): classes $\{0,1\}$ vs $\{2,3\}$

What it means — The winning split separates the two visible populations (levels 0–1 heavy, level 3 heavy) — exactly what the criterion was built to find.

Figure 1: When global fails: illumination



coins() flat vs illumination-gradient versions under three thresholding strategies. Generated by L16_thresholding_watershed.py.

How to read this figure

What to notice — On even light, Otsu ($t=107$) is clean. Add a top-to-bottom brightness ramp: fixed 128 fails outright; global Otsu misclassifies whole bands; adaptive (51 px window) recovers everywhere.

Why it happens — A single t cannot be right when a coin is darker than background elsewhere.

What it means — Global methods assume global statistics — L7's lesson, now costing segmentations.

Why it matters — Uneven light $\Rightarrow$ adaptive threshold, or normalise first (L9).

The code behind the results

```
_, bw = cv2.threshold(img, 0, 255, cv2.THRESH_BINARY+cv2.THRESH_OTSU)
dist = cv2.distanceTransform(bw, cv2.DIST_L2, 5) _, fg = cv2.threshold(dist, 0.3*dist.max(),
255, 0)
n, markers = cv2.connectedComponents(fg)
cv2.watershed(img3c, markers)
```

Parameters used — marker fraction 0.3 (sweep it!); DIST_L2. Full program: L16_thresholding_watershed.py

What it means — Four Unit-I/II tools chained: Otsu, distance (L4), components (L4), gradient flooding (L15).

Something to think about — Marker fraction 0.5? Only 2 markers survive — the sweep is mandatory.

Region-method selection

Method	Decides by	Right when	Fails when
Global/Otsu	One histogram valley	Bimodal, even illumination	Uneven light; >2 populations
Adaptive	Local neighbourhood stats	Illumination varies	Objects larger than the window
Region growing	Seed + similarity criterion	Connected, homogeneous targets	Leaks through weak boundaries
Watershed (markers)	Flooding gradient landscape	Touching convex objects	Marker extraction is fragile

Key takeaway: every method encodes an assumption about your objects — PCM-7 asks you to match assumptions to a problem, in writing.

Where this is used — your project

Segmentation lane (this IS your classical baseline) — the T8 mini-challenge and mini-project expect exactly today's chained pipeline with justified parameters and one failure case — start from the lecture code;

Change-detection lane — Otsu on the normalised difference image (as in L18's panel) is your change-mask baseline;

All lanes — whatever Unit V's networks produce, THESE are the baselines that make your improvement claims meaningful.

Try this yourself — Run the watershed recipe on one project image; sweep the marker fraction {0.2, 0.3, 0.5}; count objects at each.

Otsu's criterion, precisely

Problem: choose the threshold t that best separates the histogram into two classes — state exactly WHAT is optimised.

$$t^* = \operatorname{argmax}_t w_0(t) \cdot w_1(t) \cdot [m_0(t) - m_1(t)]^2$$

What each symbol means

- w_0, w_1 — class probabilities (histogram mass below/above t) m_0, m_1 — class mean intensities the product = between-class variance $\sigma_B^2(t)$ (argmax over 256 candidates — exhaustive, cheap)

How to read this

What it means — Otsu rewards a split whose two groups are internally tight and mutually far apart — meaningful exactly when TWO populations exist (bimodality). On a unimodal histogram the argmax still returns a t — a confident answer to a question the image never asked (the L16 sanity-check rule).

Summary

- Otsu = maximum between-class variance: optimal only where two populations exist. Uneven illumination demands local statistics or prior normalisation. Watershed + distance markers separates touching objects — its marker step deserves a parameter sweep, always.

Key formulae

$\sigma_B^2(t) = w_0 w_1 (\mu_0 - \mu_1)^2 \rightarrow \max \text{ over } t$ Markers: $\text{threshold}(\text{dist}, \alpha \cdot \max)$, $\alpha \approx 0.3$ Pipeline: Otsu $\rightarrow$ open $\rightarrow$ dist $\rightarrow$ markers $\rightarrow$ watershed

What comes next

PCM-7 (releases Fri 11 Sep, due Mon 14 Sep, ~35 min) — Segmentation design memo: one pipeline diagram for the supplied uneven-illumination, touching-objects problem — stages, parameters, expected failure.

Tuesday 15 Sep — L17 (12 PM): morphology, components, descriptors. T7 (2–3 PM): segmentation & morphology studio + PROJECT-PROPOSAL CLINIC (teams defend classical baselines).

Logistics — T7 = W7 formative checkpoint: classical pipeline design approved/narrowed. T8 (29 Sep) = the graded classical mini-challenge — two weeks out.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Which segmentation strategy?

Choose the pipeline (threshold type + extras) for each: A. Scanned text page, even lighting. B. Petri-dish colonies, touching, even lighting. C. Road cracks under tree shadows. D. Fluorescence cells on dark background, intensity varies per cell.

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, understand): State Otsu's criterion and its core assumption; give one image type that violates the assumption and the standard remedy.

Q2 (6 marks, apply): For histogram $n = [6, 1, 1, 2]$ over levels 0-3, compute σ_B^2 at $t = 0$ and $t = 2$ and state which split Otsu prefers.

Q3 (10 marks, analyse): A watershed pipeline counts 180 'cells' where a biologist counts 60; markers came from thresholding the distance transform at $0.7 \cdot \max$. Diagnose, explain the mechanism, and fix the pipeline with justification.

Further reading

GW4 Ch.10 Secs.10.3-10.5; Szeliski Ch.5; Forsyth-Ponce segmentation

Course notes: Comprehensive Notes L16; Vol. I ch. 16.

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